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Reflecting on the diversity of talks this semester

At first because the course was titled “ISA Seminar”, I came to class expecting talks more closely related to topics in information systems and applications. However, after coming to class and experiencing talks of unexpected variety, I started coming with no specific expectation about what the topic would be. After that, I felt like finding out what the topic for each week was in an unexpected manner led me to take away more from some of the talks because I would kind of just go to class with a carefree open mind, not necessarily with that kind of overly-concentrated mindset that I usually put myself into to learn about some computing-related topic or think deeply about my research progress.

Of course, I also found a lot of the computing-related topics to be very engaging as well. For example, I really enjoyed the talk about creative AI applications in music given by Yang Yi-Hsuan, because the idea of creatively using technical methods to better facilitate ways to be creative in music is very attractive to me as someone who really enjoys listening to and analyzing music. It was pleasing to find that the various projects he described all had applications of different nature. For example, he explained ways to apply machine learning to more efficiently form better mechanics-oriented tools for musicians (e.g. audio-to-lyrics alignment, singing voice separation for karaoke generation, lead sheet generation) but he also explained applications that created new opportunities for musical creativity and generation.

DJnet, the AI form of a DJ, generates music thumbnails into a mix by highlight detection and also forms comparisons with neural attention modeling of music emotion. Musical creativity or artistic expression is often regarded as something that cannot be superseded by computers, but I think one useful thing that proper applications of AI in this realm can do is to operate somewhat like an extension of the brain of the creative thinker, when directed to form certain kinds of connections they can do at a larger scale what we cannot even though we may know the prerequisite meta-knowledge about how to do so. In this sense AI at its current state can very well be used to significantly aid the creative process of music creation and other kinds of art. For example, the music medley project also explained in the same seminar uses a similarity embedding network that works better than humans for finding a suitable order of music pieces. Even if the ordering does not match the original sequence, there is a rationale behind which kinds of music “tails” match which “heads” and artists can use this medleying as a way of exploring sequence combinations they might not have considered before but could be suitable for what they are making. Even the results of projects described such as the music style transfer done by his students can generate some significant inspiration if we consider past psychological research about conceptual incongruence fostering creativity. The music style transfer serves as a form of conceptual incongruence because it *materializes* genre + composition superimpositions that may not have previously been considered and are quite obviously difficult to imagine without actually hearing.

Some less computing-related talks I enjoyed a lot were the ones given by Taitzer Wang and Aihua Lee. From Taitzer Wang the most valuable things I took away were how he was able to use his period with cancer to cultivate some very impressive skills, and with what kind of mindset he was able to do it. I am also very interested in the Taiwanese language and really appreciated his overview of the history of the Taiwanese language and its usage. Also because it was related to music theory I enjoyed Aihua Lee’s overview of her implementation for music therapy and explanation of the theory concepts behind why what kinds of sounds evoke certain feelings in us universally. That class was also generally very relaxing and pleasing to attend during what was a very stressful week for me.